

Nathan R.T. Lesnevich

PERSONAL INFORMATION

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ACADEMIC POSITIONS

2024-CURRENT Postdoctoral Fellow, Oklahoma State University

EDUCATION

MAY 2024 Ph.D in Mathematics, **Washington University in St. Louis** (WUSTL), St. Louis MO
Thesis: Splines on Cayley Graphs of the Symmetric Group.
| Advisors: Assoc. Prof. Martha Precup
| Prof. John Shareshian

MAY 2021 A.M. in Mathematics, **Washington University in St. Louis** (WUSTL), St. Louis MO

MAY 2019 B.S. In Mathematics (*magna cum laude*), **Bucknell University**, Lewisburg PA
Pure Mathematics concentration, additional fulfillment of B.A. in Computer Science.
Thesis: Positivity Among P -Partition Generating Functions
| Advisor: Prof. Peter McNamara

PUBLICATIONS

8. A. Dave, T. Feng, N. Lesnevich, S. Oh, E. Richmond, E. Wang, and A. Wu. *Shifted lower Bruhat intervals are EL-shellable*. submitted (2026) [arxiv:2608.04417]
7. J. Boretsky, A. Cornejo, R. Hodges, P. Horn, N. Lesnevich, T. McAllister. *Combarability in Bruhat orders*. submitted (2026). [arxiv:2601.21158]
6. N.R.T. Lesnevich. *Signed permutations and degree-one dot action representations for types B and C*. submitted (2025). [arXiv:2511.13913]
5. J. Lee, N. Lesnevich, and M. Precup. *Restricted inversion polynomials*. to appear in Electronic Journal of Combinatorics (2025). [arXiv:2511.05676]
4. B. Adenbaum, E. Barnard, M. Hlavacek, B. Kagy, N. Lesnevich, G. D. Nasr, and K. Waddle, *The poset of maximal tubings of the cycle graph is a lattice*. submitted (2025). [arXiv:2510.09429]
3. N.R.T Lesnevich, *Splines on Cayley graphs of the symmetric group*. Forum of Mathematics, Sigma 13 (2025): e96. [arxiv:2401.14380]
2. N.R.T. Lesnevich, *Hook-shape immanant characters from Stanley-Stembridge characters*. Algebraic Combinatorics, Volume 7 (2024) no. 1, pp. 137-157. [arXiv:2304.05285].
1. N.R.T. Lesnevich and P.R.W. McNamara, *Positivity among P -partition generating functions*. Annals of Combinatorics. 26, 171-204 (2022). [arxiv:2006.10087]

GRANTS AND FUNDING

2023 Prepared and received NSF conference grant DMS-2245927 (PI: Martha Precup)
Graduate Student Combinatorics Conference 2023
Co-written with small group of other graduate students

TEACHING

2024-CURRENT	Instructor, Oklahoma State University – <i>Calculus II (SP26)</i> – <i>Differential Equations (FL24, SP25, FL25, FL26)</i> – <i>Combinatorics (SP26)</i> – <i>Algebra (FL26)</i>
2020-2023	Graduate Teaching Assistant, Washington University in St Louis – <i>Calculus II (SP21)</i> – <i>Calculus III (FL20, SP23)</i> – <i>Differential Equations (FL21, SP22, FL22)</i>
2022-2024	Director, Washington University Graduate Reading Program
2021-2023	Instructor, Washington University Graduate Reading Program – <i>Representation Theory (FL23)</i> – <i>Lie Algebras (SP22)</i> – <i>Erhart Theory Theory (FL21)</i> – <i>Young Tableaux (SP21)</i>
2022	Summer Instructor, Washington University in St. Louis – <i>Finite Math (instructor of record)</i>

CONFERENCES AND SEMINARS

Organization

Spring 2025	Special Session on Combinatorics and Lie Theory AMS Spring Central Sectional Meeting <i>University of Kansas</i>
Spring 2023	Graduate Student Combinatorics Conference 2023 Co-wrote and received NSF conference grant DMS-2245927.
2022-2023	WUSTL Combinatorics Seminar <i>Bi-weekly seminar.</i>

Speaking

Spring 2026	The CAGE Seminar (inter-university Philadelphia) <i>GKM Theory and Reflection Groups</i> , April 16
Spring 2026	AMS Spring Southeastern Sectional, Special Session on Symmetric functions and subvarieties of flag varieties <i>Some type B/C symmetric functions from Hessenberg varieties</i> , March 29
Spring 2026	AMS Spring Southeastern Sectional, Special Session on Algebraic Combinatorics <i>The maximal tubings of the cycle graph form a lattice</i> , March 28
Fall 2025	The Schubert Seminar (inter-university) <i>Combinatorics for GKM Theory and the Dot Action</i> , Nov 3
Fall 2025	AMS Fall Central Sectional, Special Session on Interactions between Geometry, Combinatorics, and Flag Varieties <i>The Second Cohomology of $\text{Hess}(X, H)$ and the Dot Action in types B/C</i> , Oct 19

<i>Fall 2025</i>	Oklahoma State Lie Groups Seminar <i>GKM Theory and Flag Varieties</i> , Oct 8
<i>Summer 2025</i>	Graduate Research Workshop in Combinatorics, (at IA State) <i>Dihedral Splines</i> , June 15-27
<i>Spring 2025</i>	WashU Combinatorics Seminar <i>The Maximal Tubings of the Cycle Graph and G-inversions</i> , March 17
<i>Fall 2024</i>	Oklahoma State Combinatorial and Commutative Algebra Seminar <i>Splines on Cayley Graphs of the Symmetric Group</i> , Sept 26
<i>Spring 2024</i>	Poster, Enumerative and Algebraic Combinatorics (at UFL) <i>Splines on Cayley Graphs of the Symmetric Group</i> , Feb 25-7
<i>Fall 2023</i>	KU Combinatorics Seminar <i>Splines on Cayley Graphs of the Symmetric Group</i> , Nov 3
<i>Summer 2023</i>	Short Talk, Schubert Conference (at UIUC) <i>Splines on Cayley Graphs of the Symmetric Group</i> , June 20
<i>Spring 2023</i>	Poster, AlGeCom-XXIII <i>Splines on Cayley Graphs of the Symmetric Group</i> , Apr 15
<i>Spring 2023</i>	Graduate Online Combinatorics Colloquium <i>Splines on Cayley Graphs of the Symmetric Group</i> , Mar 8
<i>Spring 2023</i>	Dartmouth Combinatorics Seminar <i>Splines on Cayley Graphs of the Symmetric Group</i> , Feb 21
<i>Spring 2023</i>	WUSTL Combinatorics Seminar <i>Splines on Cayley Graphs of the Symmetric Group</i> , Feb 20
<i>Fall 2022</i>	AMS Fall Eastern Sectional, Special Session on Geometric Aspects of Algebraic Combinatorics. <i>A Proof of an Immanant Conjecture for Hook Partitions</i>
<i>Spring 2022</i>	Short Talk, Graduate Student Combinatorics Conference 2022 <i>Hook-Partition Immanant Characters from Stanley-Stembridge Characters</i>

Recent Program/Workshop Attendance

<i>Summer 2026</i>	CMND 2026 Thematic Program: Algebraic Combinatorics and Applications, at University of Notre Dame
<i>Summer 2025</i>	Graduate Research Workshop in Combinatorics, at Iowa State University
<i>Summer 2024</i>	AMS Mathematics Research Communities: Algebraic Combinatorics, at Beaver Hollow Convention Center
<i>Spring 2024</i>	BIRS Workshop: Homological Perspectives on Splines, at UBC Kelowna
<i>Summer 2023</i>	Schubert Calculus Summer School, at UIUC
<i>Spring 2023</i>	8th Lake Michigan Workshop on Combinatorics and Graph Theory, at the University of Notre Dame
<i>Spring 2022</i>	ISM Discovery School: Geometry and Combinatorics of Hessenberg Varieties, at UQAM.

PROGRAMS & OUTREACH

2024	Research advisor, Freiwald Scholars Summer Research Program, Washington University in St. Louis <i>Co-advised several undergraduate research projects</i>
2024	Assistant, Sonia Kovalevsky Day at Washington University in St. Louis <i>Cryptography Activity Organizer</i>
2021-2024	Director, Graduate Reading Program, Washington University in St. Louis <i>Recruitment, Organizational, and Administrative Duties</i>
2023	Speaker, Math Circle, Washington University in St. Louis <i>'Cryptography: The math of making and breaking codes'</i>

SCHOLARSHIPS AND CERTIFICATES

2023	Robert H. McDowell Teaching Award. <i>Department graduate teaching award, 1 annual awardee.</i>
2022	Brian Blank Teaching Award for 3rd Year Graduate Student. <i>Department graduate award, 1-2 annual awardees.</i>
2020-2021	David B. Emmes Endowed Scholarship.
2015-2019	John C. Hoover Scholarship for Students in Mathematics
2015-2019	Bucknell Presidential Fellowship <i>4 year half-tuition merit scholarship with research mentorship.</i>
2015-2019	Bucknell University's Deans List
2018	John C. Hoover Summer Research Funding Recipient
2017	Pi Mu Epsilon Mathematics Honors Society Member
2017	Bucknell Program for Undergraduate Research Award Recipient